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//2501351020 BADAL PRASAD
open System
open System.Text.RegularExpressions

// 1. Reverse a string
let reverseString (str: string) =
    str |> Seq.rev |> String.Concat

// Check palindrome
let isPalindrome (str: string) =
    let cleaned = str.Replace(" ", "").ToLower()
    cleaned = reverseString cleaned

// 2. Transform string using pattern matching
let transformString (str: string) =
    match str with
    | s when Regex.IsMatch(s, @"^[A-Za-z]+$") ->
        s.ToUpper()
    | s when Regex.IsMatch(s, @"^[0-9]+$") ->
        s + s
    | _ -> str

// 3. Validate email using Regex
let isValidEmail (email: string) =
    Regex.IsMatch(
        email,
        @"^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$"
    )
```

2501351020

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// Main program
printfn "===== STRING PROCESSING IN F# ====="

// Task 1: Reverse and palindrome
printfn "\n--- Reverse and Palindrome ---"

let words = ["madam"; "hello"; "level"]

words
|> List.iter (fun word ->
    printfn "String: %s" word
    printfn "Reversed: %s" (reverseString word)
    printfn "Palindrome: %b" (isPalindrome word)
)

// Task 2: String transformation
printfn "\n--- String Transformation ---"

let inputs = ["hello"; "1234"; "hello123"]

inputs
|> List.iter (fun s ->
    printfn "Input: %s -> Output: %s" s (transformString s)
)

// Task 3: Email validation
printfn "\n--- Email Validation ---"

let emails = [
    "student@gmail.com"
    "badal123@yahoo.com"
    "invalid-email"
    "rajveer@domain"
]
```

```
PS D:\5thSEM KMRU\NEWAGE\FSharp_Lab_Assessment\LabVI> dotnet fsi
===== STRING PROCESSING IN F# =====

--- Reverse and Palindrome ---
String: madam
Reversed: madam
Palindrome: true
String: hello
Reversed: olleh
Palindrome: false
String: level
Reversed: level
Palindrome: true

--- String Transformation ---
Input: hello -> Output: HELLO
Input: 1234 -> Output: 12341234
Input: hello123 -> Output: hello123

--- Email Validation ---
student@gmail.com : Valid Email
badal123@yahoo.com : Valid Email
invalid-email : Invalid Email
rajveer@domain : Invalid Email
PS D:\5thSEM KMRU\NEWAGE\FSharp_Lab_Assessment\LabVI>
```

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